

Note: Continuous preserves the compactness, so, a continuous map $f: X \rightarrow \mathbb{R}$ on compact X has minimum and maximum, since $f[X]$ is closed bounded.

Darboux's Theorem.

Let $f: [a, b] \rightarrow \mathbb{R}$ be a differentiable map.

Then, the derivative f' satisfies the Intermediate Value Theorem.

Proof. Suppose that $f'(a) < f'(b)$. Let α s.t. $f'(a) < \alpha < f'(b)$ be given.

Define $g(x) = f(x) - \alpha x$ ($x \in [a, b]$).

Clearly g is differentiable, and $g'(a) = f'(a) - \alpha < 0$, $g'(b) = f'(b) - \alpha > 0$.

And, since g is continuous, it has a minimum in $[a, b]$, for some c .

By the assumption, $c \neq a$ and $c \neq b$. And, since g has a minimum at c , $g'(c) = f'(c) - \alpha = 0$. So proved $\square$

Corollary. $f: I \rightarrow \mathbb{R}$ has a primitive, then $f[I]$ is interval.

Note: If f is defined on an interval and cannot satisfy IVT, then it has no primitive.